Why These Simple Laws, Forward Time, Many-Worlds, and Superdeterminism, in Brief

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Abstract

As reality is mathematical structure, derived from the pattern-randomness dichotomy (Bernstein, 2026c), algorithmic probability provides the natural measure over structure-space: P(S) proportional to 2^(-K(S)), where K is Kolmogorov complexity. Each additional bit of specification halves a structure's measure. From this single principle, convergent results follow across physics and epistemology: (1) Occam's Razor is a theorem: simpler structures are exponentially more probable; (2) the Principle of Least Action is a selection effect: variational specifications have lower K; (3) cosmological fine-tuning dissolves: the constants are among the simplest compatible with observers; (4) the arrow of time is grounded in computational irreversibility: many-to-one functions have no inverse; (5) Many-Worlds has the lowest specification cost among quantum interpretations: collapse accumulates K per measurement, MWI does not; (6) superdeterminism survives the fine-tuning objection: specificity is not complexity; (7) string theory is exponentially disfavored: 3,000+ bits of landscape selection vs a few hundred for the Standard Model; (8) simplicity preference in cognition is adaptive: organisms that prefer compressible patterns track higher-measure structures; (9) the Principle of Least Action, the arrow of time, and emergence are three instances of one structural fact: many-to-one mappings; (10) extra dimensions emerge free: the metric bundle (14D over 4D) and its infinite jet bundle tower are mathematically necessary at zero additional K, unlike string theory's 3,000+ bits. Convergence across epistemology, dynamics, cosmology, thermodynamics, computation, and quantum foundations constitutes cumulative evidence for the position.

Keywords: algorithmic probability, Kolmogorov complexity, Occam's Razor, arrow of time, superdeterminism, Many-Worlds, string theory, mathematical monism

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1. One Principle

Among all consistent mathematical structures, simpler ones are exponentially more probable. In a prefix-free coding, each program of length k occupies a fraction 2^(-k) of program-space. A structure producible by a 10-bit program has 2^10 = 1024 times the measure of one requiring a 20-bit program. Each additional bit of specification halves the fraction. This is algorithmic probability. A "program" here is a finite description, not a process requiring execution.

The structures exist as mathematics. The measure is over specifications, not computations.

The philosophical foundation is developed elsewhere: reality is identical to mathematical structure because the pattern-randomness dichotomy exhausts logical space (Bernstein, 2026c). Only consistent mathematics forms structure: inconsistent statements can be labeled but not constructed, and algorithmic probability is the unique natural measure over structure-space (Schmidhuber, 2000; Bernstein, 2026d). This paper does not re-derive the foundation. It demonstrates what algorithmic probability explains.

The argument for algorithmic probability is cumulative. Any single consequence might be coincidence. Six independent consequences from the same principle, across unrelated domains, is the kind of convergence that distinguishes a correct principle from an ad hoc hypothesis.

Some consequences follow from the foundation before any physics. The hard problem of consciousness is circular reasoning, all interactive dualisms collapse to mathematical monism, and the supernatural and libertarian free will resolve as square circles. These are developed in the companion papers (Bernstein, 2026c, 2026d). The physics follows.

2. Occam's Razor Is a Theorem

Scientists prefer simpler theories. This preference has been treated as methodological taste, aesthetic bias, or pragmatic convenience. Hossenfelder (2018) asked why physics should prefer simplicity at all, and found no satisfactory answer. Algorithmic probability provides one: lower-K structures have exponentially higher measure. The criterion is not beauty but brevity.

A theory with generating complexity K describes structures with measure proportional to 2^(−K). A competitor with complexity K > K describes structures with measure proportional to 2^(−K). The ratio is 2^(K − K), exponentially favoring the simpler theory. Occam's Razor is not a guide to inference. It is a fact about the distribution of existence.

This explains why simplicity has been so unreasonably effective as a criterion. Every major theoretical unification in physics (Maxwell unifying electricity and magnetism, Einstein unifying space and time, the Standard Model unifying three forces) reduced the total descriptive complexity of physics. Each was rewarded by the universe confirming its predictions. Algorithmic probability says this pattern will continue: observers are exponentially more likely to find themselves in structures with lower generating complexity, so the final theory of physics should be simpler than what we have, not more complex.

3. The Principle of Least Action

Classical mechanics, general relativity, quantum field theory, and the Standard Model all admit variational formulations. Systems evolve along paths extremizing the action integral. Why should nature prefer extremal paths?

Consider two ways to specify the same physics. Specification A: "systems follow paths extremizing the action functional S." Specification B: exhaustive enumeration of permitted trajectories. Specification A has vastly lower generative complexity. Under algorithmic probability, variational universes dominate the measure. Algorithmic probability does not make PoLA emerge within a universe. It selects for universes whose physics is variational, because variational specifications have lower generative complexity. The Principle of Least Action is a selection effect across structure-space, not a dynamical consequence within any particular structure. A deeper result connects PoLA to the computational arrow: both are instances of many-to-one mappings. Many paths, one classical survivor. Many inputs, one output. The Noether chain (observers → stability → conservation → symmetry → variational) forces variational physics for any observer-supporting universe. Full development is given in Bernstein (2026k).

This connects to Noether's theorem: every continuous symmetry of the action yields a conservation law. Symmetries reduce the independent specifications needed. More symmetry means lower K. Algorithmic probability predicts that fundamental physics will be maximally symmetric, which is what we observe: the Standard Model is built on gauge symmetries, and general relativity on diffeomorphism invariance.

4. Fine-Tuning Dissolves

Change the gravitational constant slightly and stars cannot form. Adjust the strong force and atoms fall apart. The universe appears calibrated for complex life.

If all consistent mathematical structures exist (which follows from the null filter argument: any filter is itself a consistent structure, so all filters exist, including none), then observer-permitting structures necessarily exist among them. We find ourselves in one because we could not find ourselves anywhere else. Selection bias, not design.

Algorithmic probability adds precision: among observer-permitting structures, simpler ones are exponentially more probable. This predicts that the fundamental constants are not arbitrary but among the simplest values compatible with observers. It also predicts that outstanding puzzles like dark matter will have simple solutions (lower-K explanations will be confirmed over higher-K alternatives), and that the initial conditions we observe are low in complexity because observers are more likely to find themselves computing from simple starting points. The same bias applies to temporal location: earlier observer-moments have shorter specification prefixes, but later moments have more total observers.

The distribution peaks where these pressures meet. Algorithmic probability also applies to competing physical theories: string theory's total specification complexity (approximately 3,000 to 4,700 bits) dwarfs any plausible generating rule for the Standard Model's 19 to 26 scalar parameters, rendering string theory exponentially disfavored. Under algorithmic probability, elegance is short description. String theory landscape-selection has so far failed its own aesthetic and parsimony criteria on the same grounds. Under mathematical monism, all string vacua exist, but low complexity of a theory itself is necessary and not sufficient: the theory must also reproduce observed physics, and no string vacuum has been shown to do so.

5. The Arrow of Time

The laws of physics are time-symmetric. Yet entropy increases. The standard approach (Albert, 2000) postulates a low-entropy initial condition, the Past Hypothesis, as a brute boundary condition. Algorithmic probability derives it.

A universe described forward from initial conditions I has total complexity K(L) + K(I). If I is low-entropy (smooth, nearly uniform), K(I) is low. The same universe described backward from final state I_f has complexity K(L) + K(I_f), where I_f encodes the accumulated output of billions of years of dynamical evolution, enormous and potentially incompressible.

Forward-described structures dominate the measure by 2^(K(I_f) − K(I)), an astronomical factor. All states exist eternally in the block universe. Observers experience the low-complexity end as the past because computation is many-to-one and consciousness can only model forward. Low entropy is low descriptive complexity. Algorithmic probability exponentially favors low complexity. The Past Hypothesis is not a brute postulate about cosmology but a consequence of what modeling means. Full development is given in Bernstein (2026e, 2026h).

The computational arrow provides the structural foundation for this asymmetry. Nearly all computations are many-to-one: 1+1=2 is deterministic, but 2=x+y is not. The inverse does not exist. This is why backward description cost is high: many-to-one functions ran forward, and any backward specification would require identifying which of the many inputs produced each output. Algorithmic probability quantifies the measure-theoretic consequence of this irreversibility. The computational arrow stands without algorithmic probability. algorithmic probability's time arrow depends on it. Deutsch's constructor theory supports the point: it defines physics through local operations, not global state descriptions. When Deutsch appeals to global unitarity to dissolve the arrow, he leaves his own construction. The arrow lives at the level constructor theory selects.

6. Superdeterminism Rehabilitated

Bell's theorem shows that no local hidden-variable theory satisfying statistical independence (SI) can reproduce quantum predictions. The mainstream dismisses superdeterminism (rejecting SI) as requiring “implausible fine-tuning” of initial conditions.

This objection conflates specificity with complexity. Initial conditions generated by simple rules have low K and are measure-favored, regardless of how specific they appear. The digits of pi are specific but not complex: a short program generates all of them.

The K-decomposition clarifies the comparison. Total specification complexity is K(L) + K(I) + K(O), where K(L) is laws, K(I) is initial conditions, and K(O) is the specification of which particular measurement outcomes occurred. Under collapse interpretations, K(O) grows without bound: each measurement selects one eigenvalue, accumulating an algorithmically incompressible string. Under both Many-Worlds and superdeterminism, K(O) = 0. No random selection occurs.

The fine-tuning objection inadvertently argues against collapse, not against superdeterminism. The Born rule, whether probabilities proportional to squared amplitudes follow from algorithmic probability or require independent derivation, remains an open problem. Many-Worlds retains a parsimony advantage with current models (the Schrödinger equation alone, no additions), but the competition is mathematical: whichever generating rule has lower total K wins.

7. Beyond Physics

The position extends beyond physics: consciousness, AI alignment, and ethics follow as structural consequences and are developed in companion papers.

8. Convergence

Five results from algorithmic probability, one independent structural fact, two from the combination:

1. Occam’s Razor (epistemology)
2. Least Action (selection across structure-space)
3. Fine-tuning (cosmology)
4. Arrow of time (thermodynamics)
5. Superdeterminism (quantum foundations)
6. Arrow of computation (structural foundation: many-to-one functions)
7. Many-Worlds strengthened (algorithmic probability + computational arrow)
8. Variational dynamics forced (many-to-one + Noether chain, Bernstein 2026k)
9. Variational geometry forced: the metric bundle (14 dimensions over 4D spacetime) emerges necessarily from any manifold with a metric, at zero additional K. String theory pays 3,000+ bits for its extra dimensions; the metric bundle pays zero. The jet bundle tower above it is infinite and

equally free. The physics is in the harmonic amplitude profile, not the dimension count

10. String theory's vibrational spectrum redeemed as derived geometric overtones rather than postulated objects (Bernstein 2026n)
11. Encoding hierarchy: Turing mapping preserves data, the holographic principle preserves physics within entropy bounds, jet bundle reconstruction preserves geometry exactly: three distinct senses of encoding suggesting a deeper structural pattern

Each was previously treated as a separate puzzle. Mathematical monism unifies them. The computational arrow is the structural foundation. Algorithmic probability quantifies its consequence. Together they converge on a single account.

The principle makes a meta-prediction: future puzzles in fundamental physics will have simple solutions. Complexity is the exception, not the rule. If the pattern holds, algorithmic probability is confirmed. If a future discovery requires irreducibly high-K explanation, algorithmic probability is challenged. The principle is falsifiable through its consequences.

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All companion papers available at ORCID: 0009-0009-1761-2867